

Stat 3113: Lesson 1

Introductory Material Part 1

Learning Objectives

- The topics covered here concern methods of drawing conclusion from the data. These methods form the field of inferential statistics, also called statistical inference.
- But before we discuss these topics, we must first learn more about methods of collecting and summarizing clearly the basic information they contain.
- These topics of sampling and descriptive statistics will be covered in the rest of this chapter.

Definitions

- **Population:** is the **entire collection** of objects or outcomes (elements) about which information is sought.
 - The **number** of elements in the population is denoted by **N**.
- **Sample:** is a **subset of a population**, containing the objects or outcomes that are actually observed.
 - **Sample size:** is the **number** of elements in the sample. It's typically denoted by **n**.
- **Observation:** is the **value**, at a particular **period**, of a particular **characteristic**.



Example:

Suppose that there exists 7 million college students in the United States today. And let's assume the average GPA of all these college students is 3.0. If we select 150 college students to compute the average GPA.

1. What is the population?

The 7 million college student in the United States today.

2. What is the sample?

The 150 selected college students.

3. What is the observation?

The GPA of a selected college student

4. Find N and n ?

$N \rightarrow 7$ million

$n \rightarrow 150$

Why sample?



Save resources: time, money, workload.

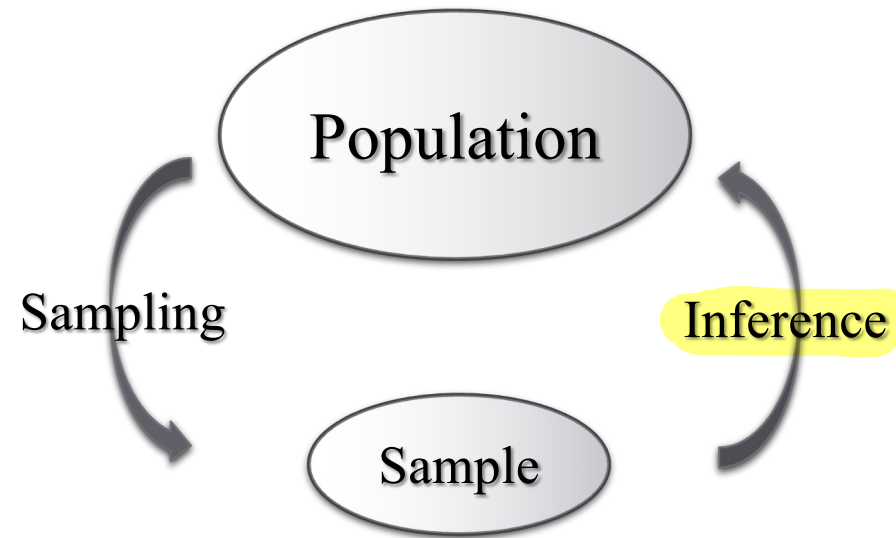


The elements in the population can't be identified.



Destroying the population elements (observations).

- US population Estimated at 334,233,854 on Jan 1, 2023
 - This represents an increase of $\approx 0.47\%$, from New Year's Day 2022.
- From United States Census Bureau.



Why sample?



Save resources: time, money, workload.



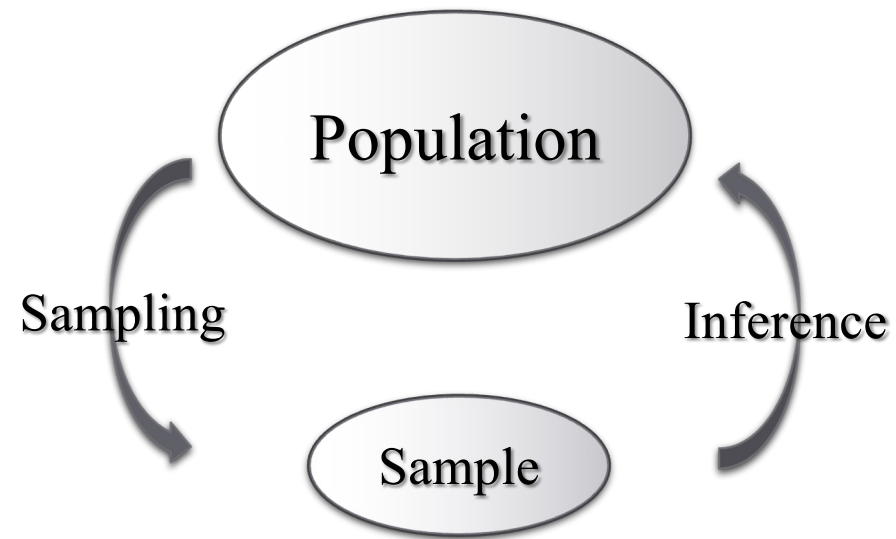
The elements in the population can't be identified.



Destroying the population elements (observations).

Example

- Grains of salt or sugar ①
- Stars in the sky
- Sand on the beach
- Grains of Rice in a Bowl
- A basket of socks. ②



Why sample?



Save resources: time, money, workload.

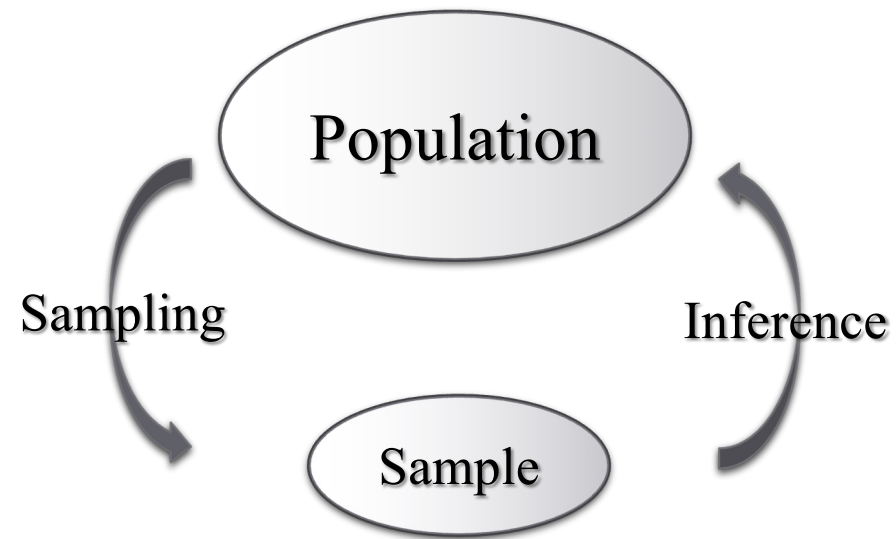


The elements in the population can't be identified.



Destroying the population elements (observations).

Example : Component Testing / Material strength Testing.
• In engineering, materials are often tested for their strength and properties. If you're testing a material, by subjecting it to extreme forces, temperature changes or chemical reactions, the material might become compromised and break during the testing, making it impossible to gather further data.



Representative Samples and Bias in Sample

- The sample must contain the characteristics of the population in order to be a **representative sample**.
- In statistics, a **sampling bias** is created when a sample is collected from a population and some **members** of the population are **not as likely to be chosen as others** (remember, each member of the population should have an equally likely chance of being chosen).
 - Examples: - Health habits of students at Missouri S&T
- Internet Usage Online Survey / User Feedback in Product development.
- When a **sampling bias** happens, there can be **incorrect conclusions** drawn about the population that is being studied.



- **Simple random sample (SRS)**: is a sample chosen by a method in which each collection of the n population items is equally likely to make up the sample, just as a lottery.
- A SRS is not guaranteed to reflect the population perfectly.
 - SRS's always differ in some ways from each other; occasionally a sample is substantially different from the population.
 - Two different samples from the same population will vary from each other as well.
- This phenomenon is known as **sampling variation**.

- **Sample of convenience** is a sample that is obtained in some convenient way, and **not drawn by a well-defined random method.**

Things to consider with convenience samples:

- Differ systematically in some way from the population.
- **Only used when it is not feasible to draw a random sample.**

No Feasible Cases:

- Population characteristics: Studying a rare species of animal (Red wolf, Philippine crocodile)
- Limited Accessibility: If the target population is geographically dispersed or difficult to access. (Europe, Africa, Asia, etc)
- Ethical Constraints: The study involve vulnerable populations (Victims of domestic violence or Abuse) or sensitive topic (LGBTQ+, Drug use, prostitution, etc.).

Examples

1. A utility company wants to conduct a survey to measure the satisfaction level of its customers in a certain town. There are 10,000 customers in the town, and utility employees want to draw a sample of size 200 to interview over the telephone. They obtain a list of all 10,000 customers, and number them from 1 to 10,000. They use a computer random number generator to generate 200 random integers between 1 and 10,000 and then telephone the customers who correspond to those numbers. **Is this a simple random sample?**

• Yes, because a random number generator was used to obtain the 200 customers making up the sample.

1. A quality engineer wants to inspect electronic microcircuits in order to obtain information on the proportion that are defective. She decides to draw a sample of 100 circuits from a day's production. Each hour for 5 hours, she takes the 20 most recently produced circuits and tests them. **Is this a simple random sample?**

• No, because the quality engineer has only sample the 20 most recently produced circuits, which exclude the principle of randomness in favor of a convenient approach.

Variables and Data

- A **variable** is a characteristic that **changes** or **varies over time and/or for different individuals** or objects under consideration.
- **Examples:** Hair color, white blood cell count, time to failure of a computer component, height, weight, etc.

How many variables have you measured?



Univariate data: One variable is measured on a single individual.



Bivariate data: Two variables are measured on a single individual.



Multivariate data: More than two variables are measured on a single individual.

Types of Variables

- **Qualitative/Categorical variables** measure a quality or characteristic on each individual or unit.

- **Examples:**

- Hair color (black, brown, blonde...)
- Make of car (Dodge, Honda, Ford...)
- Gender (male, female)
- State of birth (California, Arizona,....)

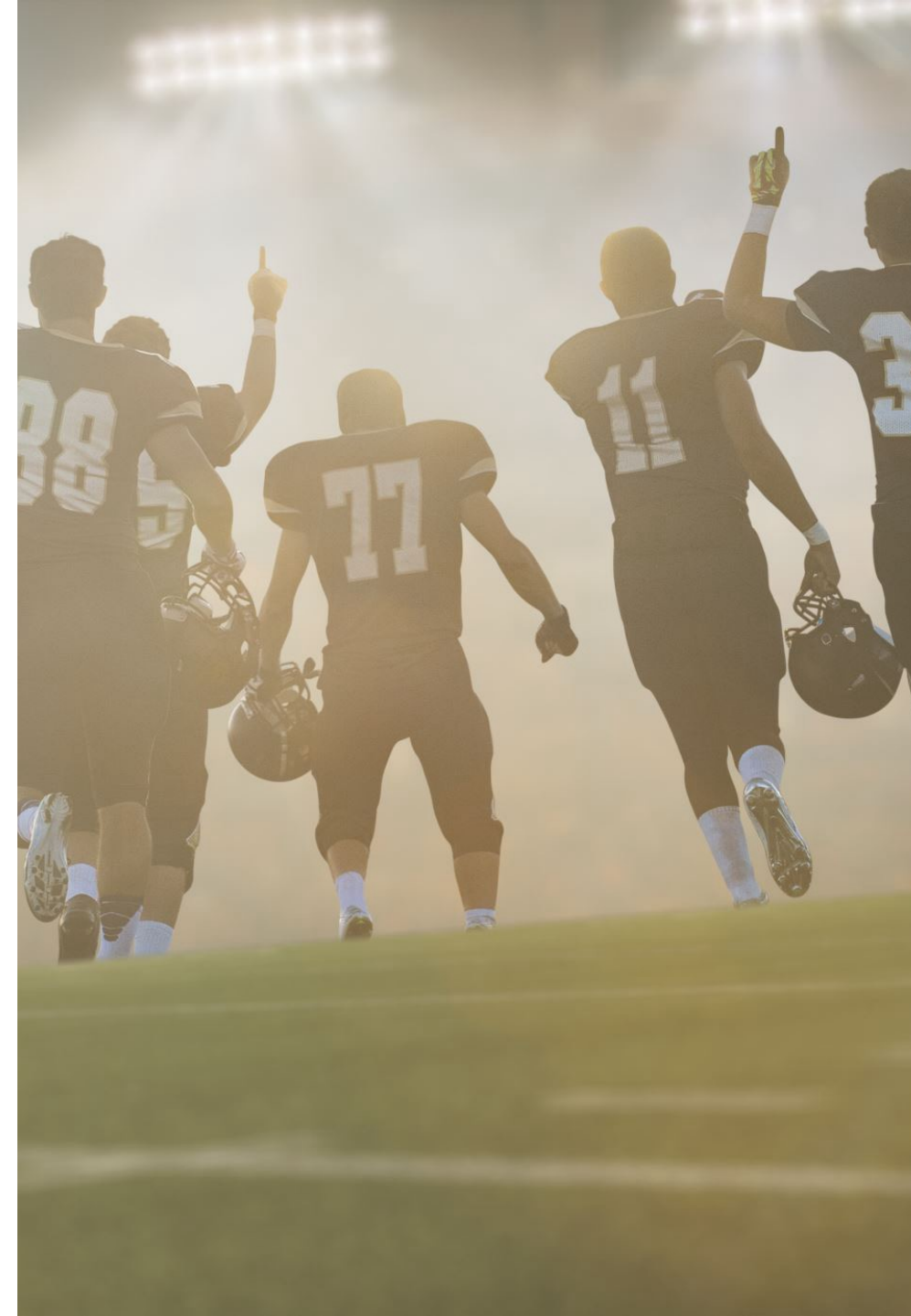
Nominal Variable

- Kind of categorical variable
- The values “name” the attribute uniquely.
- The value does not imply any ordering of the cases, for example, jersey numbers in football.
- Even though player 32 has higher number than player 19, you can't say from the data that he's greater than or more than the other.

↑ the number worn on a player's uniform.

More Examples:

- Material Types: “steel”, “aluminum”, “plastic”, “ceramic”.
- Engineering disciplines: “Mechanical”, “Electrical”, “chemical”.
- Coding standards in Software engineering: “C++”, “Python”, “Java”





Ordinal Variable

- Kind of categorical variable
- When attributes can be rank-ordered...
- Distances between attributes do not have any meaning, for example, code Educational Attainment as 0=less than H.S.; 1=some H.S.; 2=H.S. degree; 3=some college; 4=college degree; 5=post college
- Is the distance from 0 to 1 the same as 3 to 4?

Types of Variables

- **Quantitative variables** measure a **numerical quantity** on each individual.

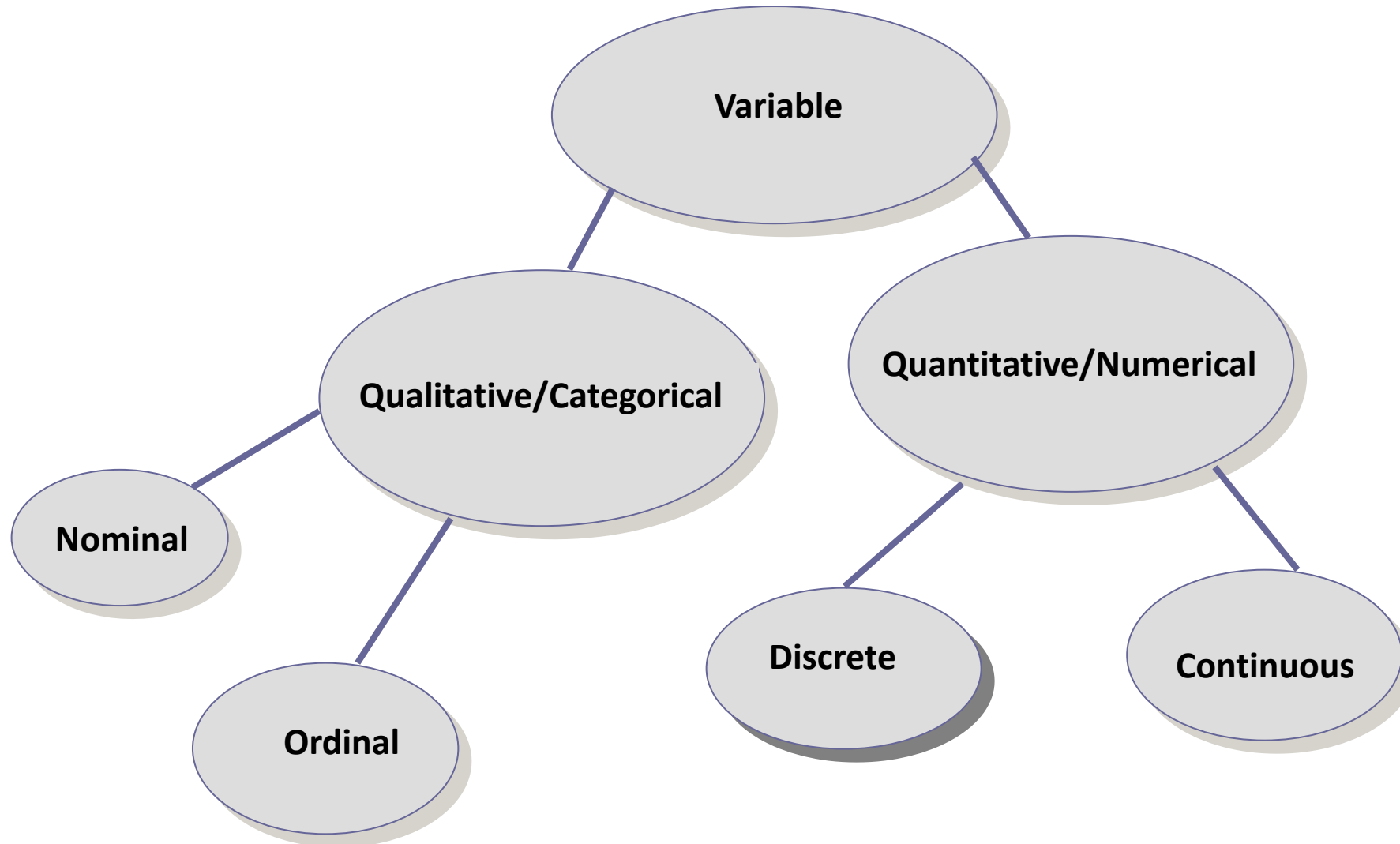
- ✓ **Discrete** if it can assume only a **finite or countable** number of values.

*Ex :- Number of defects in products on a production line.
- Number of student in this class! 30!*

- ✓ **Continuous** if it can assume the **infinitely many** values corresponding to the points on a **line interval**.

*Ex :- Time to failure of a Computer component!
- There is an infinite number of possible values between any two given temperatures.*

Types of Variables



Reading

Chapter 1 – Section 1.1



Next Topic



Introductory Material Part 2



Chapter 9 – Section 9.1